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CLOGGED SPACES, UNENDING CASES: A QUALITATIVE INQUIRY ON PNP'S IMPOUNDED VEHICLE DISPOSAL ISSUES

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the Study

Impounded vehicles, encompassing cars, motorcycles, and other motorized units seized by law enforcement for traffic violations, criminal involvement, or disputed ownership,represent more than just confiscated property. In the context of this research,they symbolize the accumulated weight of unresolved legal issues, procedural delays, and administrative inefficiencies plaguing the Philippine National Police (PNP). (Agaton et al., 2020) Emphasized that unmanaged transport assets contribute to urban pollution and inefficient land use. Without clear disposal protocols, these vehicles become liabilities. Addressing this issue requires coordinated reform across law enforcement, judiciary, and asset management agencies. While existing regulations aim to standardize procedures through clear auction rules and inter-agency coordination, discrepancies persist between policy and practice, particularly in resource-constrained regions (Badua, 2021).

In the United States, state laws, municipal ordinances, and private towing company practices collectively govern vehicle seizure, notification, and disposal. This fragmented regulatory landscape has led to persistent issues such as predatory towing, inconsistent recordkeeping, and hardships for low-income vehicle owners who often rely on their cars for housing (Kumar, 2021). Consequently, some states have revised policies to incorporate procedural safeguards,reduce impoundments for minor infractions, and improve owner notification systems (Klosky & Lerman, 2024). Moreover,the transportation sector in the U.S. has emphasized the adverse effects of prolonged impoundment on freight carriers, with excessive

fees and cargo delays prompting calls for federal oversight (Indarsih, Y. 2021). Environmental concerns further complicate disposal, as abandoned vehicles can leak harmful substances like battery acid and oil into the soil and water, necessitating costly cleanups (Aborisade & Ariyo, 2022). These experiences in the U.S. illustrate that impounded vehicle backlogs can persist even with robust infrastructure and administrative systems when legal complexities, inadequate oversight, and environmental risks converge.

Parallel patterns emerge in the Philippines, although the administrative, legal, and logistical contexts differ significantly. National regulations, including updated Land Transportation Office memorandum circulars from 2021 onwards, outline standardized procedures for inventorying, notifying owners, appraising, and auctioning unclaimed impounded vehicles (Calabias, 2022). The Philippine National Police has issued operational guidelines for the safekeeping and disposal of seized vehicles, and periodic inventory audits are mandated to ensure accountability (Aborisade & Ariyo, 2022). However, compliance varies, especially at the local level where coordination between the PNP,the LTO, prosecutors, and local governments is paramount.

Locally, the Commission on Audit (2022) reported that PNP stations in Metro Manila and the Caraga Region have turned into vehicle graveyards, obstructing public access and compromising operational safety. The Land Transport Authority (2025) mandates the disposal of unclaimed vehicles after six months; however, implementation remains inconsistent. Agaton et al. (2020) emphasized that unmanaged transport assets contribute to urban pollution and inefficient land use. ULAP (2023) highlighted weak coordination between LGUs and national agencies as a barrier to enforcement. The PNP Mobility Asset Guidelines (2023) emphasize the importance of

periodic inventory and disposal, but compliance varies. These findings reveal a systemic failure in treating impounded vehicles as time-sensitive assets that require inter-agency action.

The accumulation of vehicles in storage consumes limited space, diverts administrative resources away from core policing functions, and imposes unnecessary financial burdens on public agencies (Sang, M. 2024). This study aims to investigate the challenges and coping strategies of stakeholders involved in managing impounded vehicles within the Philippine National Police, focusing on the lived experiences of frontline officers, vehicle owners, local governments, and regulatory bodies. The primary objectives are to identify the operational, legal, and logistical issues encountered in handling impounded vehicles;to describe the strategies – both formal policies and improvised practices – employed by impounding authorities to address these challenges; and to propose practical, policy-relevant solutions for improving disposal processes. Such solutions may include harmonized inter-agency protocols, digitized inventory systems, streamlined notification procedures, and scheduled public auctions that are legally compliant and publicly transparent (Talha et al., 2023). The ultimate goal is not only to address immediate operational inefficiencies but also to bolster public confidence in law enforcement property management.

The significance of this study lies in its potential to inform policy and operational reforms. Documenting the experiences of police officers, local officials, and affected communities, it offers grounded insights into restructuring impoundment practices. The LTO (2022) emphasized the need for transparent and efficient disposal protocols. ULAP (2023) highlighted the role of institutional coordination in service delivery. The research contributes to public asset management

and environmental governance. Ultimately, it aims to transform impounded vehicles from stagnant liabilities into resolved cases and reclaimed public spaces.

1.2 Review of Related Literature

This section synthesizes existing research,relevant policies, and established best practices related to the management and disposal of impounded vehicles. By establishing the theoretical and empirical groundwork for this study, it identifies key gaps and provides insights that inform the overall research direction.

Stakeholder Challenges in Impounded Vehicle Management

Legal Ambiguities and Ownership Conflicts. Legal ambiguity in the disposal of impounded vehicles arises from fragmented mandates among the PNP, LGUs, and the judiciary. Brillantes et al. (2025) Emphasized that institutional fragmentation in Philippine governance leads to overlapping responsibilities and administrative paralysis. It is evident in impound cases where no single agency has apparent authority to initiate disposal. The absence of harmonized legal triggers results in prolonged custody and asset deterioration. In a related study, Mendoza et al. (2020) analyzed coordination failures in Philippine law enforcement and found that unclear jurisdictional boundaries often stall operational decisions. They recommend formal inter-agency agreements and disposal protocols to streamline the turnover of assets. Without such reforms, impounded vehicles remain in legal limbo, contributing to congestion and inefficiency. These findings underscore the need for legal coherence and institutional integration in the management of public assets.

Adding to these problems are bureaucratic delays in releasing, disposing of, or auctioning impounded vehicles. These delays often stem from incomplete documentation, overlapping legal jurisdictions, and poor coordination among government entities. Heeks et al. (2021) argued that these procedural complexities cause vehicles to remain impounded for extended periods, contributing to congestion and limiting capacity for new confiscations. This bureaucratic inertia inflates storage and security costs and perpetuates a cycle of inefficiency that hinders effective enforcement of vehicle laws. Consequently, impound facilities can become long-term storage sites rather than temporary holding areas, undermining law enforcement objectives.

Furthermore,the accumulation of impounded vehicles presents significant health and environmental hazards, especially in densely populated urban areas. Changed (2021), emphasized that abandoned and neglected vehicles often leak toxic substances like oil,fuel, and coolant, contaminating nearby soil and water. These pollutants can cause significant environmental damage, posing risks to public health and biodiversity. In some cases,the improper disposal of hazardous fluids and vehicle parts violates environmental regulations, highlighting a disconnect between policy and implementation in public sector waste management. Collaboration between environmental agencies and law enforcement is crucial to ensure compliance with safety and ecological standards.

Finally, a lack of standardized impound management protocols further complicates logistical challenges. Varied systems of record-keeping, maintenance, and disposal across different regions and police jurisdictions result in inconsistent practices and accountability issues. (Badua, 2021) and (Callistus, 2025) have pointed to the absence of uniform guidelines as a significant contributor to inefficiency, with some facilities conducting regular audits while others

rely on outdated manual tracking systems. These inconsistencies undermine transparency and make it difficult to assess national performance in impound management. A nationwide framework mandating uniform procedures,technological integration, and performance monitoring could alleviate these logistical barriers and promote operational consistency across law enforcement units.

Insufficient Budget Allocation. Budget allocation has consistently posed challenges for the Philippine National Police (PNP), particularly in addressing logistical concerns, such as the disposal of impounded vehicles reported. The 2025 General Appropriations Act imposed significant cuts on PNP information technology projects, reducing allocations from ₱600 million to ₱232 million. Such reductions illustrate how operational priorities are compromised when funds are insufficient. This underfunding directly impacts the PNP's ability to maintain and dispose of seized vehicles, resulting in overcrowded police lots and operational inefficiencies (Gascon, 2025).

Chronic underfunding significantly hinders the effective management of vehicle impound facilities, posing logistical and infrastructural challenges for law enforcement agencies, preventing them from adequately maintaining, securing, and modernizing their operations. Velazques (2025) emphasized that limited budgetary allocations lead to facility deterioration, staffing shortages, and a lack of essential maintenance tools and security systems. Impound operations often receive insufficient resources because available funds are prioritized for immediate policing needs like crime prevention and community engagement. This imbalance can result in unsecured facilities, increased vulnerability to theft or vandalism, and compromised record-keeping, ultimately undermining public trust in institutional efficiency and transparency. Even well-designed management frameworks struggle without adequate financial support.

The absence of a standardized national budgetary policy or financial framework for impound operations further contributes to disparities in service quality and enforcement. 11 noted that regions with stronger internal revenues or local government support tend to have more efficient impound facilities compared to those in less developed areas. This disparity results in unequal treatment and inefficiencies within the judicial process, as vehicle owners in resource-poor regions face longer retrieval times and increased bureaucratic hurdles. Furthermore, a lack of centralized funding impedes equitable distribution of financial support for facility upgrades,technology adoption, and staff training. Establishing a uniform budgeting system under the Department of the Interior and Local Government (DILG) or the Philippine National Police (PNP) could promote consistent quality and accountability nationwide.

Limited funding also impedes the integration of essential digital infrastructure. (Senješ et al., 2024) observed that the absence of a unified national database for impounded vehicles leads to fragmented and inefficient monitoring systems reliant on manual record-keeping. This technological deficiency delays case resolutions and increases the risk of administrative errors and data manipulation. Instances of missing vehicles due to inconsistencies between paper records and actual inventory underscore the pressing need for a digitized tracking system. Adequate budget allocation would enable the development of secure, interconnected databases to track vehicles from confiscation to final disposition, enhancing operational transparency and inter-agency coordination.

Finally, the sustainability of impound operations requires continuous investment in both human and physical resources, areas frequently neglected due to competing financial priorities. Research by Mahoro (2021) indicates that insufficient training and compensation for personnel

assigned to impound facilities contribute to low morale and reduced efficiency, exacerbating management difficulties. A lack of funding for regular maintenance,facility improvements, and technology procurement reinforces a cycle of inefficiency that hinders long-term reform. Institutionalizing dedicated budget lines for impound management and capacity-building initiatives can strengthen both the operational resilience and ethical governance of the system.

Judicial backlogs. Judicial backlogs significantly delay the release or disposal of impounded vehicles in the Philippines. Because vehicles classified as evidence in criminal cases, their release requires prosecutorial clearance or a court order. This dependency on judicial timelines means that congested dockets and deferred hearings stall police asset management.

Caparas & Feliciano (2025) In their article "The Problem of Delay in the Philippine Court System," published in the Philippine Law Journal, they argue that docket congestion gravely undermines the administration of justice. They note that delays in case processing are "seriously disabling" and affect all facets of the judicial process, including evidentiary custody. It directly relates to impounded vehicles, which remain in police custody until courts act on motions for release or forfeiture.

In an article for the Philippine Daily Inquirer, Cabato (2024) emphasized how new anti-colorum regulations demand a court order before confiscated vehicles can be released. Even administrative seizures by the Land Transportation Office (LTO) cannot be established without recourse to the courts, which increases reliance on legal procedures. It reinforces the inertia that prolongs vehicle custody, Cabato emphasizes, adding to the burden already placed on courts by traffic.

Judicial Reforms' Effects (2021). Judicial Reforms to Address Court Congestion and Delays in Justice in the Philippines detailed how the implementation of reforms, such as continuous trial guidelines and electronic case management, helped reduce backlogs. Congestion still exists, though, and evidence assets, such as seized cars, remain susceptible to extended detention due to court deadlines.

Accountability Gaps and Asset Deterioration. Accountability gaps in impound management often arise from unclear custodial responsibilities and weak documentation. Vehicles seized in police operations. For extended periods, yet monitoring systems remain inconsistent. (Grace & Tiempo, 2025) stated that Passive Voice in Police Reports: Balancing Objectivity and Accountability, observed that reporting practices often obscure responsibility by depersonalizing agents. This bureaucratic opacity mirrors impound operations, where unclear custodial attribution leads to lapses in accountability and public suspicion of mismanagement.

Asset deterioration is a direct consequence of prolonged storage without proper safeguards in place. PNP Memorandum Circular No. 2025-039 provides guidelines for the disposition of lost, stolen, damaged, or destroyed property, emphasizing relief from accountability when assets deteriorate beyond recovery. Applied to impounded vehicles, the absence of preventive maintenance and protective storage accelerates decay. Oil leaks, rusting, and battery corrosion compromise asset value, making eventual disposal less economically viable. Communities near impound sites also bear environmental risks, reinforcing the need for facility upgrades.

The economic implications of asset deterioration are significant. Vehicles that could have been auctioned or returned to owners lose market value due to neglect. Magno-Ballesteros et al.

(2023) argued in "Through the Lens of Accountability: Body-Worn Cameras in Police Operations" that accountability mechanisms improve institutional performance when compliance is enforced. His findings suggest that similar accountability frameworks—such as transparent asset tracking—could mitigate deterioration by ensuring custodial responsibility. Without such systems, prolonged custody translates into lost opportunities for restitution and revenue generation.

Accountability gaps also reflect broader governance challenges. Based on Vicente (2023) study on police misconduct and accountability in administrative cases, she highlighted that weak enforcement of accountability rules leads to recurring lapses in transparency. In the Philippine context, impound operations perceived as opaque or prone to corruption erode public trust. Citizens often suspect unequal treatment in vehicle release, with influential individuals securing faster clearance while ordinary motorists endure prolonged delays. Such perceptions underscore how lapses in accountability in asset management erode institutional legitimacy.

The combined effect of accountability gaps and asset deterioration is both operational and reputational in nature. Impound yards become symbols of inefficiency, with deteriorating vehicles serving as visible reminders of unresolved cases and weak custodial practices. Environmental hazards compound the problem, creating liabilities for police units and local governments. Addressing these issues requires standardized custodial protocols, environmental safeguards, and transparent reporting systems. By closing accountability gaps and preventing asset deterioration, impound operations can shift from being liabilities to becoming models of institutional integrity and efficiency.

Financial Disputes Financial disagreements among law enforcement,towing companies, and private contractors frequently hinder the effective administration of impounded vehicles. Irfan (2023) pointed out that disputes over service fees, storage expenses, and contractual duties often delay the release or proper disposal of these vehicles,resulting in administrative bottlenecks that worsen existing backlogs in impound facilities. These conflicts generally stem from unclear pricing and inconsistent billing,fostering mistrust among stakeholders. The resulting financial strain not only impacts the operational efficiency of towing and impound services but also generates public frustration, as vehicle owners are frequently caught in the middle of unresolved financial negotiations. Furthermore, such disputes consume valuable time and resources that could be better allocated to enhancing service delivery and facility management.

A significant factor contributing to these financial disagreements is the absence of well-defined contractual agreements and standardized service frameworks. As (Ifeanyichukwu et al. (2025) noted, many public-private partnerships in impound management operate under loosely structured contracts that fail to clearly specify responsibilities, payment terms, or performance metrics. This ambiguity allows for conflicting interpretations of obligations and promotes inefficiency. The problem is compounded by inconsistent government oversight, where weak enforcement of contract provisions allows disputes to persist without timely resolution. These delays disrupt the smooth operation of impound facilities and damage relationships between the public and private sectors, highlighting the urgent need for contract reform and standardized operating procedures in vehicle impoundment services.

Moreover, unresolved financial disputes often escalate into legal conflicts, prolonging the impoundment process and increasing operational costs. When service providers and law

enforcement agencies resort to litigation to resolve payment disagreements, vehicles remain in limbo, stored indefinitely pending court decisions. Irfan (2023) emphasized that these prolonged legal proceedings result in both financial losses and the deterioration of seized assets,further complicating their eventual disposal or auction. This lengthy resolution process reflects systemic inefficiencies in managing contractual relationships and underscores the importance of establishing alternative dispute resolution mechanisms, such as mediation or arbitration,to address conflicts more efficiently. These mechanisms could significantly reduce costs and expedite the return or liquidation of impounded vehicles, ultimately improving overall system efficiency.

From a public image perspective,the private sector often bears the brunt of public dissatisfaction arising from delayed or mishandled impoundment cases. Movchan et al. (2021) observed that vehicle owners frequently associate negative impound experiences with both government authorities and their private contractors, leading to reduced trust in public service delivery. This reputational damage can discourage reputable companies from participating in public contracts, limiting the pool of qualified service providers. Implementing transparent pricing,regular audits, and clearly defined performance standards can mitigate these challenges and strengthen collaboration between public institutions and private contractors. By addressing financial disputes through improved governance and accountability, impound management can evolve into a more efficient and credible public service system.

Storage Congestion and Environmental Risks. Storage congestion in vehicle impound yards presents both logistical and environmental challenges, especially in urban areas. In a study on urban infrastructure and pollution, Reyes et al. (2021) found that unmanaged accumulation of

vehicles and waste materials contributes to land degradation and inefficient space utilization. Although their research focused on solid waste, the findings apply to impounded cars that leak oil, coolant, and battery acid into unsealed ground. These pollutants pose long-term risks to soil and groundwater, particularly during the rainy season when runoff spreads contaminants. (Coracero et al., 2021) reviewed solid waste management practices and emphasized that public storage facilities often lack containment infrastructure, leading to unmanaged runoff and ecological degradation. He recommends integrating environmental safeguards such as impermeable flooring and runoff control systems. Without proper disposal timelines and infrastructure, impound yards become long-term ecological liabilities. These findings underscore the need for coordinated environmental audits and facility upgrades in impound operations.

The insufficient integration of sustainability principles into law enforcement operations is evident in the disregard for environmental management standards. Mo (2023) suggests that environmental management is often secondary to storage capacity and administrative efficiency in impound operations. This leads to inadequate waste disposal systems, a lack of spill containment infrastructure, and limited environmental safety awareness among personnel. Without proper training and resources, impound sites continue to be pollution sources rather than controlled storage environments. However, adopting green practices, such as installing oil-water separators, implementing recycling programs, and developing eco-friendly infrastructure, could significantly reduce their environmental impact and align operations with national environmental goals.

The long-term environmental consequences of neglected impound sites extend beyond localized pollution. Welsh Carroll et al., (2023) emphasize that hazardous substances such as oil,

brake fluid, and battery acid from abandoned vehicles can seep into the soil, contaminating agricultural lands and water systems. These contaminants disrupt aquatic ecosystems, diminish soil fertility, and create ongoing health risks for humans and wildlife. The cumulative effect of this environmental degradation means that even small-scale negligence can have lasting ecological impacts. Moreover,the decay of vehicle materials, including plastics and metals, contributes to microplastic accumulation and the release of volatile organic compounds, degrading air quality and increasing greenhouse gas emissions.

Beyond chemical pollution,the accumulation of vehicle debris and waste materials at impound sites creates further environmental hazards. Farooq (2021) observed that decomposing vehicle components, like tires and upholstery, attract pests and rodents, contributing to public sanitation problems. The burning or improper disposal of these materials releases toxic gases that worsen local air pollution. Together, chemical and physical degradation transform impound facilities into environmental hotspots instead of secure storage locations.

Absence of adequate training. A significant impediment to effective and ethical vehicle impound management is the insufficient training of PNP personnel and other involved parties. As Deng et al., (2024) point out, a lack of specialized knowledge in asset preservation, legal compliance, and evidence management can lead to mishandling of vehicles. This may result in damage, loss, or even contamination of crucial evidence, potentially undermining legal proceedings and exposing agencies to accusations of negligence or corruption. Without adequate technical skills, personnel may struggle to implement appropriate security and maintenance procedures, increasing the risk of theft, vandalism, or environmental problems. Therefore,

comprehensive training programs are crucial to ensuring that everyone involved in impound management possesses the necessary skills to meet professional and legal standards.

Research on policing consistently shows that specialized training positively affects operational efficiency and asset management in law enforcement. Stanton et al., (2022) discovered that training initiatives focused on asset preservation, accurate documentation, and procedural compliance substantially improve institutional performance and public accountability. These programs empower personnel to responsibly manage impounded assets, ensuring compliance with all judicial and administrative requirements. Conversely, a lack of such training contributes to procedural errors, miscommunication, and inconsistent practices across different areas. A standardized national training curriculum for impound management could address this knowledge gap and promote greater consistency in law enforcement nationwide.

The issue is further complicated by the limited availability of ongoing professional development. Shamsi et al. (2022) noted that many law enforcement agencies do not offer regular refresher courses or workshops on emerging best practices, updated laws, or relevant technological tools for impound operations. This stagnation leaves personnel unprepared to handle evolving challenges such as digital record-keeping, environmental regulations, or advanced security systems. This lack of professional development can also negatively impact morale, as employees who don't receive training or recognition for skill improvement may be less motivated to perform efficiently or uphold institutional standards. Investment in continuous learning programs is essential to ensure that impound management personnel stay informed and adaptable in a rapidly changing operational landscape.

Financial constraints. Financial limitations remain a significant obstacle to the effective and sustainable operation of vehicle impoundment systems. The substantial costs involved in securing, maintaining, and disposing of impounded vehicles, especially those left unclaimed for long periods, place a heavy burden on resources. For the Philippine National Police (PNP), these expenses often surpass allocated budgets, diverting crucial funds from essential activities like patrol services, crime prevention, and investigations (Vitiello et al., 2023). This financial pressure underscores a systemic budgetary imbalance within public institutions, where resource allocation involves continuous negotiation to balance competing priorities. Consequently, impound management can become a lower priority, negatively affecting the quality and security of stored assets.

This persistent lack of consistent funding disrupts long-term planning and hinders operational efficiency. Inadequate financial resources prevent investment in suitable storage facilities, surveillance systems, and maintenance equipment necessary for vehicle preservation and accountability (Suryanarayana et al., 2023) The absence of this infrastructure not only leads to vehicle deterioration but also increases the risk of theft and loss of crucial evidence in criminal cases. As studies have demonstrated, chronic underfunding of asset management systems often results in diminished service quality, workflow inefficiencies, and reduced institutional credibility (Moran, 2023) Addressing these financial constraints requires budgetary adjustments and policy reforms that prioritize sustainable asset management practices.

These financial challenges extend beyond government agencies to private contractors and towing companies partnering with the PNP for vehicle storage. They also face operational challenges when payments are delayed insufficient. These businesses must maintain facilities,

compensate staff, and secure insurance while awaiting government reimbursements or vehicle owners' payments (Moran, 2023) Protracted compensation delays threaten their financial stability, potentially leading to service interruptions and increased tensions between public and private stakeholders. Similar patterns occur in other public-private partnerships, where a lack of financial transparency and delayed payments impede continuous service delivery and harmonious stakeholder relationships (Shapovalov & Diachenko, 2023).

Ultimately, these financial constraints undermine the overall accountability and effectiveness of vehicle impoundment systems. When agencies and contractors operate under financial pressure, maintenance standards suffer,transparency is compromised, and public dissatisfaction increases. This creates a cycle of inefficiency and mistrust, as citizens perceive the system as unreliable or exploitative. Vitiello et al., (2023) Ensuring adequate funding, establishing timely reimbursement processes, and implementing performance-based budgeting could improve financial governance and promote operational sustainability. Effective resource management not only ensures proper asset handling but also reinforces public trust in law enforcement institutions responsible for upholding fairness and accountability (Wyllie, 2021).

Bureaucratic Inefficiencies. Bureaucratic Inefficiencies. Bureaucratic inefficiencies critically impede the smooth operation of vehicle impoundment and retrieval. Protracted administrative processes, overlapping roles, and poor communication between involved parties frequently delay vehicle release,frustrating both staff and the public (Welsh Carroll et al., 2023). Procedural bottlenecks often arise from unclear lines of authority spanning the PNP, local government, and private impound contractors. This division leads to inconsistent policy enforcement, redundant paperwork, and uncertainty about which body is ultimately responsible for releasing vehicles

(Chenzi et al., 2023). These problems contribute to a sense of disorganization and erode public faith in institutional competence.

Furthermore, these inefficiencies are often worsened by a lack of standardized documentation and digital solutions in managing impounded vehicles. Relying on manual record-keeping and outdated tracking prolongs administrative verification, increasing the risk of misplaced files and procedural errors (Wyllie, 2021). The absence of streamlined data-sharing hampers coordination, as each entity may maintain its own separate and potentially inconsistent database. Research in public sector management has consistently demonstrated that digital transformation and automation can significantly reduce bureaucratic delays, improve transparency, and strengthen accountability (Shamsi et al., 2022). Consequently, modernizing administrative procedures through information technology is vital for addressing inefficiencies and improving public service.

Additionally, inter-agency coordination remains a persistent barrier to the efficient disposal of impounded vehicles in the Philippines. Diokno (2023) examined the implementation of the Strategic Trade Management Act and highlighted how fragmented agency mandates and a lack of shared digital systems delay operational decisions across enforcement bodies. Although focused on customs, her findings reflect broader coordination issues among the PNP, LGUs, and judiciary in asset disposal. Ballesteros et al., (2023) studied inter-LGU cooperation in waste management and found that shared infrastructure and joint protocols significantly reduced service delays and improved accountability. Their model offers a replicable framework for impound operations, where overlapping mandates often stall disposal timelines. Both studies emphasize the importance of formal agreements, shared databases, and synchronized workflows to reduce

bureaucratic inertia. Without these reforms, impounded vehicles remain in custody for extended periods, contributing to congestion and asset deterioration. These findings underscore the need for institutional integration and cooperative governance in impound management.

Finally, the persistence of these inefficiencies undermines broader communityoriented policing and administrative reform goals. When the public consistently experiences slow and disorganized impound services,their confidence in the justice system declines, hindering cooperative crime prevention efforts (Chenzi et al., 2023). Institutional inefficiency becomes a sign of governance failure, highlighting the need for structural reforms that promote inter-agency collaboration, clear accountability, and public accessibility. By simplifying procedures, adopting transparent digital systems, and improving staff training the PNP and its partners can cultivate a culture of efficiency and responsiveness. Efficient bureaucratic processes not only improve impound operations but also strengthen the legitimacy of law enforcement in the public eye (Wyllie, 2021).

Strategies adopted by Impounding Authorities in Coping with Operational and Logistical Challenges

Space Optimization and Facility Management Recognizing the increasing pressure on impound facilities, authorities are exploring space optimization strategies to address operational bottlenecks. The rising number of vehicles impounded due to enhanced traffic law enforcement and road safety initiatives has strained existing storage capacities. To mitigate congestion and improve storage efficiency, solutions such as vertical parking systems,rotational retrieval protocols, and collaborations with private storage companies have been implemented (Waheeb, 2023). Research demonstrates that these spatial management innovations can enhance storage

efficiency by as much as 30% in densely populated impound environments (Edwards et al., 2022). These strategies not only alleviate physical congestion but also improve operational throughput,facilitating the timely and organized storage and retrieval of vehicles.

Furthermore, effective space optimization enables better monitoring of impounded vehicles,reducing instances of prolonged storage and vehicle deterioration. Implementing systematic parking arrangements and scheduled retrieval processes can decrease clutter and minimize safety risks within impound facilities (Waheeb, 2023). Efficient space management also improves staff performance, enabling officers to locate, process, and release vehicles more quickly, leading to improved service and greater public satisfaction. Studies suggest that structured storage systems within law enforcement logistics significantly reduce processing errors and operational delays, which are frequently observed in congested impound lots (Edwards et al., 2022).

Digitization and Technological Integration. The shift towards digitized records and computerized inventory management is now crucial for improving the traceability and accountability of impounded vehicles. Linking vehicle information with national registration databases not only minimizes errors and accelerates processing times but also fosters greater transparency (Dambazau, 2022). Research indicates that automated tracking systems can significantly decrease case backlogs and resolve retrieval disputes, with observed reductions of up to 40% within two years of implementation (Cordner et al., 2025). This technological integration enables authorities to effectively track impound durations, prioritize vehicles for

disposal or release, and mitigate asset deterioration,thereby strengthening both procedural efficiency and legal compliance.

Digital inventory systems are essential in modernizing police logistics and impound management. Relatedly, Pontanar (2025) highlighted in a study on digital transformation challenges in Police Regional Office 9, that manual systems remain prone to inefficiencies and accountability lapses. He argued that adopting digital technologies enhances transparency and responsiveness in law enforcement operations. Applied to impound yards, digital inventory reduces paperwork, strengthens audit trails, and accelerates the release or disposal of items. These innovations directly address congestion by streamlining custody records.

Digital tracking also improves chain-of-custody visibility, a critical factor in evidentiary management. Mercado (2024) emphasized in a study on accountability mechanisms in Pampanga police operations that technology-driven monitoring, such as body-worn cameras and digital logs, strengthens institutional performance. Applied to impound yards, similar systems ensure that vehicles classified as evidence remain secure while awaiting judicial clearance. Real-time monitoring also helps detect irregular movements, reinforcing accountability. Such safeguards are vital in preventing pilferage or unauthorized release.

Community trust fortified through digital transparency. Villafuerte (2025) emphasized that weak accountability mechanisms in local law enforcement erode public confidence. Digital inventory systems counter this by providing clear, accessible records of impound operations. When communities can verify custody and release processes, perceptions of corruption or favoritism

diminish. Transparency through digital platforms thus becomes a governance tool, aligning enforcement with public expectations of fairness. This integration of technology into impound management enhances both efficiency and legitimacy.

Digital inventory systems are essential for facilitating broader institutional reforms. Research indicates that work ethics and accountability structures influence police performance, as well as digital compliance systems (Claire Tindoy & Magbojos, 2025). When applied to impound yards, integrated platforms ensure that custodial practices align with evolving legal requirements. This comprehensive approach transforms impound management from a potential liability into a model of institutional integrity. By linking impound records with judicial and transportation databases, agencies can better coordinate timelines for release, forfeiture, or disposal, ultimately reducing congestion and preventing asset deterioration. This holistic approach transforms impound management from a liability into a model of institutional integrity. By linking impound records with judicial and transport databases, agencies can synchronize timelines for release, forfeiture, or disposal, reducing congestion and asset deterioration.

Collaborative Partnerships. To effectively manage operational and logistical challenges, strategic alliances with local government units (LGUs), towing services, and accredited auction houses are now critical. Multi-party agreements facilitate a more decentralized impoundment system, increased storage space, and more efficient vehicle disposal, all in accordance with laws like Republic Act No. 10586 (Semenyshyn et al., 2023). Partnering with towing companies ensures vehicles are moved quickly from where they were seized to impound facilities, which alleviates congestion and operational delays (Farooq, 2021).

Working with accredited auctioneers is also key to legally disposing of unclaimed vehicles, creating revenue and freeing up valuable storage space (Sonde et al., 2024). Involving LGUs in creating satellite impound yards helps distribute the workload and makes services more accessible, especially in densely populated cities. Research indicates that when various agencies collaborate in public service logistics, it leads to greater efficiency, less duplication of effort, and better adherence to legal and environmental regulations (Semenyshyn et al., 2023).

Personnel Training and Capacity Building. Effective operations in impounding services heavily rely on comprehensive training and professional development for personnel. Specialized programs that emphasize safe vehicle handling, accurate documentation, environmental responsibility, and ethical customer service significantly improve both technical skills and interpersonal abilities (Yu, 2021). Research suggests that well-trained staff are more likely to follow established procedures, which in turn leads to a decrease in disagreements and complaints from vehicle owners (Fourchard, 2023).

Furthermore, continuous capacity-building initiatives equip staff to effectively use new technologies, digital platforms, and navigate evolving legal landscapes. This ensures sustained high performance and the ability to address new challenges. Blomley et al., (2022) argue that investing in personnel strengthens an organization's reputation and reduces errors in impounding processes. Training programs that utilize scenario-based learning,role-playing exercises, and mentorship opportunities demonstrably improve decision-making abilities and workplace efficiency, leading to both improved operational quality and greater public satisfaction.

Environmental Management and Sustainability. Environmental responsibility is now a key part of how vehicle impound facilities operate. When vehicles can't be reclaimed, authorities are

using environmentally sound disposal techniques, like certified dismantling and recycling, adhering to environmental regulations (Rand et al., 2024). These efforts help lower the amount of waste sent to landfills and lessen the chance of soil and water pollution from dangerous fluids like oil, brake fluid, and battery acid (Fomenko et al., 2024).

What's more, incorporating sustainable practices builds public confidence and strengthens the credibility of these institutions. By reclaiming valuable materials for reuse and sticking to environmental standards, impounding operations show they're committed to social responsibility and align with principles of environmentally conscious governance (Rand et al., 2024). Studies indicate that environmentally responsible disposal can cut ecological hazards by as much as 50%, ensuring that law enforcement activities contribute to both public service and sustainability objectives (Fomenko et al., 2024).

Financial and Budgetary Strategies. Maintaining effective operations within tight budgets calls for creative financial planning. Cost-recovery approaches, such as storage fees, auction revenues, and penalty charges, help augment limited government resources (Yu, 2021). Authorities can also optimize budgets through strategies like bulk purchasing, shared maintenance agreements, upgrades to more energy-efficient facilities, allowing them to maximize resource use while preserving service quality (Kaleem, 2022).

Financial stability is especially important for high-capacity impound facilities, where insufficient funding can result in operational failures and potential legal issues. As (Yu, 2021) points out, good financial management ensures continuous service delivery, prompt maintenance, and adherence to legal requirements. Furthermore, incorporating sound budget

planning and monitoring practices enables authorities to allocate resources strategically, lessening the risks linked to both facility overcrowding and resource scarcity.

Policy-Relevant Innovations for Improving Impounded Vehicle Disposal Processes

Digital Inventory Management and Data Integration. The adoption of digital inventory systems has significantly reshaped impounding operations,resolving many of the problems inherent in manual documentation and vehicle tracking. Ezeji (2024) notes that traditional paper-based systems are often susceptible to errors,redundancy, and lost records, which in turn lengthen retrieval and disposal processes. To mitigate these issues, law enforcement agencies are increasingly implementing automated tracking systems that provide real-time vehicle status and location updates. Callistus (2025) found that these systems substantially decrease the time required for impound verification by directly linking vehicle records to national registration databases and judicial portals. This facilitates improved inter-agency collaboration, leading to faster, more accountable decision-making. (Zsigmond & Kovács, 2021) also emphasize that this digital integration enhances operational transparency and helps prevent unauthorized vehicle releases and manipulation of records.

Furthermore,this digitization is consistent with broader public governance efforts to improve efficiency and promote data-driven decision-making. The integration of these systems into national digital infrastructure supports the Philippine government's e-governance initiatives, designed to reduce bureaucratic processes and enhance transparency (Mahoro, 2021). Cordner et al., (2025) reported that jurisdictions utilizing automated impound systems have seen a 40% reduction in administrative backlogs and disputes related to vehicle retrieval. These findings highlight the ways in which digital technologies can streamline impound management,reduce

human error, and strengthen public confidence in law enforcement. By creating secure,traceable, and interconnected data platforms, authorities can better ensure that impounding procedures adhere to accountability standards, while also expediting the legal disposal of vehicles.

Public Auction Modernization and Legal Streamlining. Modernizing public auction procedures offers a promising solution for clearing impound backlogs and boosting government revenue. (Kamchatov et al., 2021) demonstrated that transparent and regularly scheduled auctions foster bidder confidence, which, in turn, increases participation and improves recovery rates. Mellijor (2022) underscored the critical role of online auction systems, or e-auctions, particularly during pandemic-related restrictions, as they maintained disposal activities while broadening access to potential buyers. Furthermore, digital auction platforms provide audit trails that help prevent corruption, ensuring fairness and adherence to government procurement regulations. These advancements highlight technology's potential to democratize access and enhance the financial sustainability of impound operations.

Complementing these auction reforms, streamlining legal processes addresses procedural inefficiencies that often delay vehicle disposal. Pedere et al., (2025) contend that minimizing documentary requirements and standardizing clearance forms promote administrative consistency while upholding due process. Simplified protocols also curtail opportunities for corruption by reducing discretionary power in release and disposal decisions. Callistus, (2025) further observes that integrated auction and legal frameworks facilitate a more rapid transition from impoundment to disposal, optimizing storage space and lowering costs. This synergy between auction modernization and legal standardization ultimately fosters a more transparent, accountable, and revenue-efficient impound system.

Green Disposal Practices and Environmental Compliance. Impounded vehicles can leak oil, fuel, and battery acid, contaminating soil and waterways. Barreda (2022) found that weak enforcement of RA 9003 at the LGU level in Camarines Sur led to improper handling of hazardous waste. It mirrors risks in police impound facilities, where prolonged storage without safeguards can worsen liabilities. Green impound protocols are necessary to align custodial operations with environmental standards.

Furthermore, the Ecological Solid Waste Management Act (RA 9003, amended 2018) provides the national mandate for segregation and safe disposal of hazardous materials. The Extended Producers' Responsibility Act (RA 11898, 2022) further strengthens accountability by requiring manufacturers and custodial facilities to ensure proper end-of-life management of vehicles and parts. These laws establish the legal foundation for green impound protocols (RA 9003, 2018; RA 11898, 2022).

Manabat et al., (2024) in their study of Candaba, Pampanga, they demonstrated that impounding systems have a significant impact on both ecosystems and communities. Environmental protections must be incorporated effectively into daily operations, including leak checks, segregation, and coordination with accredited DENR transporters. This case demonstrates how localized practices can inform police impound protocols.

Additionally, DENR Administrative Order 2021-07, which outlines the Guidelines for the Environmentally Sound Management of Waste Electrical and Electronic Equipment (WEEE),

emphasizes the importance of accredited handlers and standardized disposal procedures. Although focused on electronics, its principles apply to impounded vehicles and hazardous components. Benchmarking international protocols, Philippine reforms emphasize environmental screening at intake and pre-disposal stages.

The local experiences of Camarines Sur and Pampanga demonstrate that green impound protocols are not only a national mandate but a regional necessity. Local initiatives complement national laws, such as RA 9003 and RA 11898, creating a framework for sustainable custodial management. Integrating environmental safeguards into daily operations reduces risks and strengthens institutional credibility. Ultimately, green impound practices align police operations with both local ecological governance and national policy frameworks (Barreda, 2022); (Paul Manabat et al., n.d.).

Cross-Agency Collaboration and Technological Transparency. Cross-agency collaboration has been emphasized in recent PNP directives to address the backlog of impounded vehicles. Memorandum Circular No. 2025-049 prescribes standardized procedures for reporting, custody, and disposal, clarifying the roles of PNP, LTO, and LGUs in vehicle management. This alignment reduces task duplication and strengthens accountability among agencies. By embedding responsibilities into SOPs, agencies minimize procedural delays and ensure lawful timelines for disposal. Such collaboration directly reduces congestion in police impound yards. It also enhances institutional trust by demonstrating coordinated governance (Philippine National Police, 2025; Land Transportation Office 2022.).

Technological transparency has emerged as a governance lever in impound management. The PNP Highway Patrol Group (HPG) is finalizing its Information Management System (IMS) to modernize law enforcement operations. This digital platform integrates vehicle data, enabling officers to verify the status of stolen or impounded cars instantly. Audit trails and electronic logs deter manipulation and loss of records, strengthening oversight. Transparency through technology also enhances public confidence in law enforcement. Citizens gain assurance that vehicles are not subject to arbitrary disposal. Thus, technology strengthens accountability while reducing opacity in operational processes (Caliwan 2022; Pontanar, 2025).

Local-level reforms demonstrate how technology and collaboration converge to improve impound management. LGUs working with PNP units have piloted uniform templates for intake documentation and digital reporting. These practices ensure that vehicle status updates are accessible to both agencies and courts. Regular joint reviews between the Philippine National Police (PNP) and the Land Transportation Office (LTO) are effective measures for preventing case delays. By ensuring transparency during the intake process, these agencies minimize the chances of informal negotiations or the withholding of documents. Such reforms directly tackle issues like yard congestion and asset deterioration. They also emphasize the need to scale technological solutions to help resource-limited local commands (Paris, 2025; Land Transportation Office, 2022).

Despite these advances, challenges persist in achieving interoperability and ensuring public transparency. System integration across PNP, LTO, and civil registries is uneven, limiting the full potential of digital reforms. Empirical evidence on disposal turnaround times remains scarce, requiring further study. Moreover, public dashboards or portals for impounded vehicle status are

not yet standardized. This gap reduces citizen access to timely information and weakens participatory oversight. Future research should evaluate the impact of digital adoption on disposal timelines. It should also explore how transparency tools can be made a standard practice within institutions, across agencies. Addressing these gaps will strengthen accountability and reduce the perception of "unending cases" (Pontanar, 2025).

In summary, managing impounded vehicles within the Philippine National Police (PNP) and among other stakeholders faces a variety of complex challenges, as demonstrated by the review of related literature. The main problems are fragmented record-keeping systems, bureaucratic delays, inadequate budget allocation, and overcrowded and underdeveloped impound facilities. In addition to jeopardizing asset preservation and evidence integrity, these logistical and infrastructure issues also raise operating costs and reduce the effectiveness of law enforcement. Furthermore, neglected vehicles pose a threat to public health and violate ecological standards due to environmental hazards, including pest infestations and toxic leaks. The absence of standardized protocols and digital infrastructure exacerbates inconsistent practices and disparities in accountability among regions.

In addition, judicial backlogs, corruption, financial disputes, and communication breakdowns all contribute to systemic inefficiencies. Prolonged Vehicle impoundment damages public trust and strains institutional resources due to delayed case resolutions and inadequate interagency coordination. The misuse of seized property enabled by corruption and lax oversight disrupts operations and harms reputations due to unresolved financial disputes between law enforcement and private contractors. It highlighted the need for sustainable practices due to the environmental degradation caused by hazardous materials and improper waste management. To

enhance operational resilience, ethical standards, and public trust in impound management,the literature ultimately advocates for integrated reforms, including unified management frameworks, digital systems, and transparent governance.

1.3 Theoretical Framework

This study is firmly rooted in Institutional Theory (Meyer et al., 2021) and Accountability Theory (Romzek & Dubnick, 1987). These two theories provide the conceptual framework for our analysis of the challenges and coping strategies in managing and disposing of impounded vehicles within the Philippine National Police (PNP).

Institutional Theory

Suggests that organizations often adopt formal structures, rules, and procedures to gain legitimacy. However, in the context of impounded vehicle disposal, this Theory helps to explain the persistent gap between national-level policies and the inconsistent compliance observed at local police stations. The accumulation of vehicles in storage yards is a clear manifestation of how institutional pressures and procedural complexities hinder the effective implementation of standardized disposal protocols, despite their existence (Meyer et al., 2021).

Accountability Theory

Emphasizes the importance of custodial responsibility, transparency, and accountability in public institutions. (Romzek & Dubnick, 1987) identify multiple dimensions of accountability—bureaucratic, professional, political, and legal—that shape organizational performance. Applied to impound management, this theory highlights how unclear custodial responsibilities, weak

documentation, and lapses in oversight contribute to asset deterioration, public suspicion of mismanagement, and erosion of trust in law enforcement. By framing impounded vehicles as public assets, Accountability Theory underscores the need for transparent reporting systems, standardized custodial protocols, and mechanisms that ensure equitable treatment of stakeholders.

Together, these theories support the study by linking stakeholder experiences (frontline officers, vehicle owners, local governments, and regulatory bodies) with systemic governance challenges. The practical implications of our findings could lead to significant improvements in impound management. Institutional Theory explains the persistence of clogged impound yards despite formal disposal policies, while Accountability Theory highlights the consequences of weak custodial practices and transparency gaps. Anchoring the research in these frameworks enables a deeper understanding of how organizational structures and accountability mechanisms influence the operational, legal, and logistical issues surrounding the disposal of impounded vehicles.

1.4 Schematic Diagram

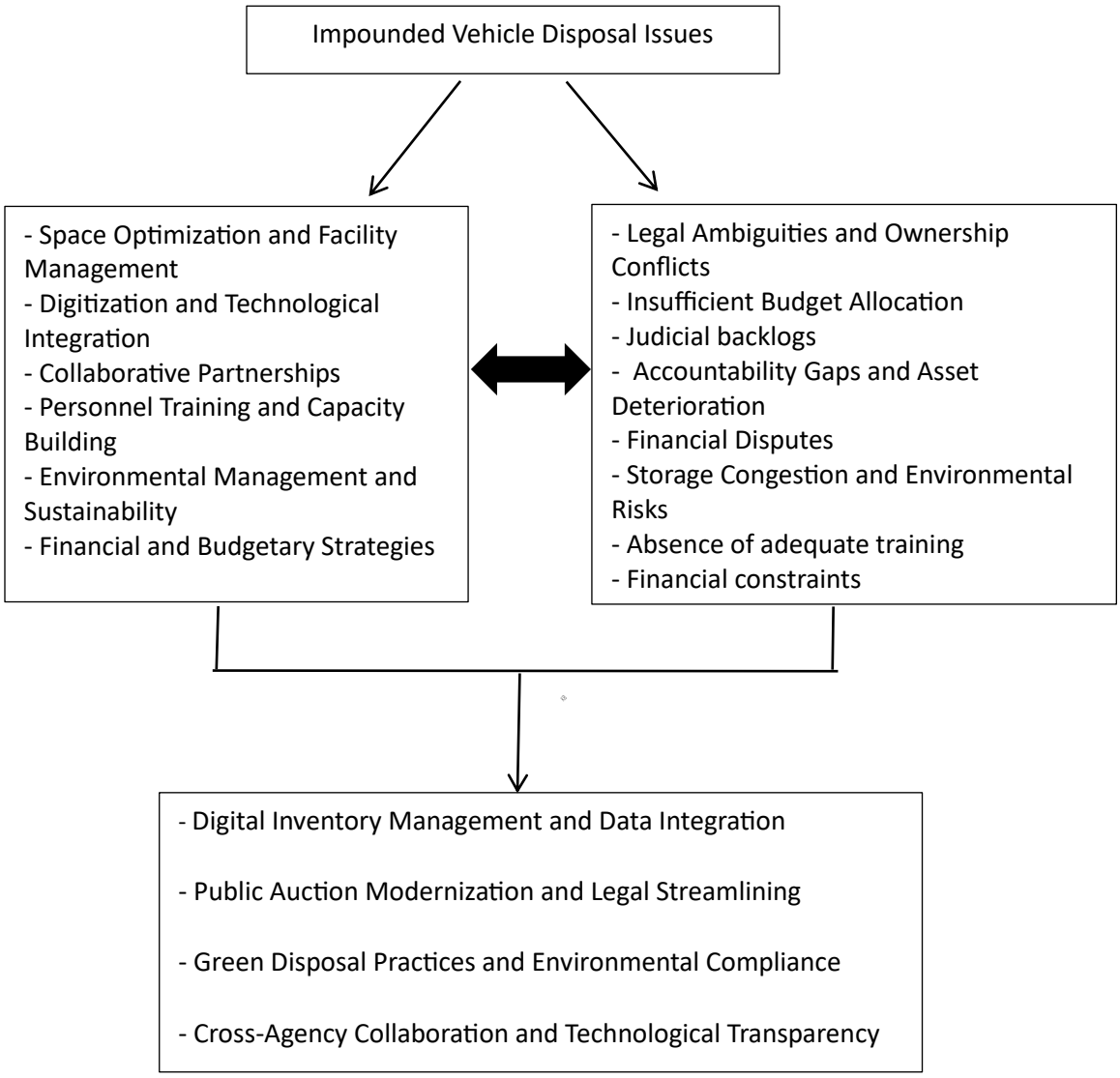


Figure 1. The Theoretical Framework of the Study

1.5 Objectives of the Study

This study has a clear focus on examining the challenges faced by stakeholders in managing impounded vehicles within the Philippine National Police, a crucial aspect of law enforcement and public safety. The specific objectives are:

1. To identify the problems experienced by the PNP and other stakeholders in handling impounded vehicles.

2. To describe the strategies adopted by impounding authorities in coping with operational and logistical challenges.

3. To recommend grounded, policy-relevant innovations for improving impounded vehicle disposal processes.

1.6 Definition of Terms

The following terms are operationally defined to enhance the understanding of the study.

Absence of Adequate Training – Refers to insufficient education and hands-on instruction provided to impounding personnel, resulting in limited knowledge of operational procedures, safety protocols, and administrative processes. This study addresses the inadequate skill development of officers responsible for handling vehicle documentation, storage, and disposal, which impacts efficiency and compliance with regulations.

Bureaucratic Inefficiencies – Denotes delays and procedural complexities caused by excessive administrative layers and unclear lines of authority. Impounded vehicle management refers to the slow processing of vehicle releases and disposal clearances, often due to redundant paperwork and inadequate inter-agency coordination.

Communication Breakdowns – Refers to failures or delays in transmitting critical information among law enforcement agencies, local government units, and other stakeholders. This study addresses misunderstandings or coordination lapses that result in errors in documentation, retrieval delays, or the mismanagement of impounded vehicles.

Coping Mechanisms – refer to the strategies, methods, and adaptive responses employed by impounding authorities and stakeholders to address operational and logistical challenges in handling impounded vehicles, as observed within the Philippine National Police (PNP) context.

Corruption and Lack of Accountability – Describes unethical practices and misuse of authority, such as bribery, favoritism, or manipulation of records, which compromise transparency and fairness. In the context of impounding operations, this refers to the absence of strict monitoring systems and sanctions that allow irregular releases or auction process irregularities. **Environmental Concerns** – Encompasses the ecological implications of vehicle impoundment, such as oil leakage, rusting, and improper disposal of hazardous materials. In the study, this term refers to the need for sustainable and eco-friendly practices in handling abandoned or non-retrievable vehicles to prevent soil and water contamination.

Financial Constraints – Refers to the limitation of available funds to support operational requirements, maintenance, and infrastructure development. In this context, it highlights how insufficient financial resources hinder the implementation of technological upgrades, facility expansion, and personnel training programs.

Impounded Vehicles – refer to motor vehicles seized and kept under the custody of the PNP or other authorized agencies due to legal violations, pending investigation, or unresolved ownership disputes.

Insufficient Budget Allocation – Pertains to the inadequate distribution of government funds to impounding units, which restricts their capacity to maintain facilities, procure equipment, or digitize operations. These highlight the constraints that hinder the prevention of effective implementation of policy-driven reforms.

Judicial Backlogs – Refers to the accumulation of unresolved legal cases that delay the release or disposal of impounded vehicles. In the study, 'therefore' refers to the impact of slow judicial processing on prolonged storage, resulting in congestion and increased maintenance costs.

Logistical and Infrastructural Problems – Encompasses challenges related to space management, facility design, transportation, and storage capabilities. In this context, it describes the shortage of properly equipped impounding yards, inadequate towing vehicles, and a lack of systematic retrieval mechanisms that hinder operational efficiency.

Policy-Relevant Innovations – refer to proposed strategies, systems, or procedural reforms derived from research findings that aim to improve impounded vehicle disposal processes' efficiency, transparency, and sustainability

Problems Experienced – refer to the operational, logistical, administrative, and legal issues encountered by the PNP and other stakeholders in the management, storage, and disposal of impounded vehicles.

Stakeholders – individuals, groups, or organizations directly or indirectly involved in or affected by impound management processes, including the PNP, towing service providers, local government units, vehicle owners, and the judiciary.

Strategies Adopted - section refers to the actions, measures, and approaches impounding authorities take to address challenges such as storage limitations, procedural delays, and administrative inefficiencies in impounded vehicle handling.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter provides a detailed account of the research setting,the study population, and the sampling methods employed to ensure the findings are derived from reliable and representative sources. It also outlines the data analysis plan, explaining how the collected information will be interpreted in order to address the study's aims. Finally,the chapter emphasizes the rigorous ethical considerations implemented to maintain integrity, protect confidentiality, and guarantee respect for all participants involved in the research.

2.1 Research Design

This study will adopt a descriptive-qualitative research design to explore the experiences of the Philippine National Police (PNP) and other relevant stakeholders in managing impounded vehicles. A descriptive-qualitative approach is appropriate for studies that aim to provide straightforward, detailed accounts of participants' perspectives and practices, without the deeper philosophical commitments of phenomenology (Villamin et al., 2025).

The design emphasizes a rich description of lived reality, focusing on how stakeholders articulate the operational and logistical challenges they encounter and the coping strategies they employ to address them. By centering on participant narratives, the study seeks to generate comprehensive, contextualized insights into current practices and highlight potential areas for improvement in managing law enforcement assets.

2.2 Research Materials and Methods

The primary data collection method will consist of semi-structured interviews guided by a rigorously developed questionnaire. The use of the tool to gather detailed responses about the challenges faced and strategies used in managing impounded vehicles. To ensure clarity, relevance, and alignment with the research objectives, the questionnaire will undergo expert validation before administration. After validation, interviews will be conducted individually, either in person or via online platforms, depending on participant availability and accessibility. With informed consent, all interviews will be audio-recorded and transcribed directly to ensure the accuracy and completeness of the data.

In alignment with the descriptive-qualitative orientation of this study, the data will be analyzed using qualitative data analysis software, such as NVivo or MAXQDA. These tools will facilitate the systematic collection of the data before thematic analysis, enabling efficient organization, retrieval, and transparency through the analytical process. During the coding stage, labels will be assigned to significant statements and responses, forming the basis for identifying broader patterns.

2.3 Research Environment, Population, and Sample

This study will be conducted in selected municipalities, specifically Buenavista and Nasipit Agusan del Norte. These sites offer a pertinent setting as Philippine National Police (PNP) personnel in these areas are actively involved in managing impounded vehicles. Participants will be the fifteen (15) PNP officers directly responsible for the impounding, safeguarding, and disposal of vehicles. To ensure the collection of rich, qualitative data relevant to the research

objectives, purposive sampling will be used to identify fifteen participants with direct, relevant experience in impounded vehicle disposal within the Philippine National Police (PNP). To ensure the depth and credibility of insights, the following criteria will guide participant selection:

- **Affiliation**: Must be active PNP personnel assigned to units or departments directly involved in the management, safekeeping, or disposal of impounded vehicles (e.g., Evidence Custodian, Logistics, Traffic Enforcement, or Investigation).
- **Experience**: Must have at least three (3) years of continuous service in roles related to impound operations, disposal procedures, or asset management.
- **Knowledge Scope**: Should demonstrate familiarity with operational protocols, legal procedures, and inter-agency coordination relevant to vehicle impoundment and disposal.
- **Willingness to Participate**: Must voluntarily consent to participate in the study and be willing to share their professional experiences and perspectives.

This targeted selection ensures that the data gathered reflect informed, practical viewpoints from personnel with firsthand experience of the systemic challenges and coping mechanisms in impound management.

2.4 Research Data and Analysis Plan

This qualitative study will employ semi-structured interviews guided by a comprehensive questionnaire to elicit detailed narratives concerning operational challenges and coping strategies. Open-ended questions will encourage participants to respond in their own words, generating descriptive accounts of their lived experiences.

All interviews will be audio-recorded and transcribed verbatim. To ensure systematic and rigorous analysis, the transcribed data will be processed using qualitative data analysis software such as NVivo or MAXQDA. These tools will facilitate efficient coding, data organization, and retrieval, thereby enhancing transparency, consistency, and analytical depth.

The data analysis will follow (Braun & Clarke, 2006) the six-phase framework for thematic analysis, situated within a descriptive-qualitative orientation:

1. **Familiarization with the Data** – Immersing in the transcripts through repeated reading and noting initial observations.
2. **Generating Initial Codes** – Systematically identifying and labeling significant statements relevant to the research objectives. Coding will precede thematic analysis to ensure that raw data are meaningfully categorized.
3. **Searching for Themes** – Clustering codes into potential themes that reflect underlying issues and coping strategies.
4. **Reviewing Themes** – Refining and validating themes to ensure coherence and alignment with the data corpus.
5. **Defining and Naming Themes** – Articulating the essence of each theme and its relevance to the research questions.
6. **Producing the Report** – Constructing a coherent narrative grounded in the data that addresses the study's objectives.

Through this methodology, this study aims to develop a nuanced understanding of the challenges stakeholders face and the coping mechanisms they employ to manage impounded

vehicles. The use of qualitative software and library-based literature will ensure analytical rigor and academic integrity.

2.5 Research Ethical Standards

The conduct of this study will strictly adhere to established research ethics standards, ensuring the protection of the rights, dignity, and welfare of all participants, while maintaining the study's trustworthiness and integrity.

A. Intellectual Property Rights (IPR)

This study recognizes and respects intellectual property rights by properly acknowledging all creative works, ideas, and research findings. Researchers provide due credit to authors, scholars, and institutions whose theories, tools, and materials informed the development of this work. As noted, respecting intellectual property is crucial for establishing the credibility of research by confirming its originality and authenticity. To that end, we adhere to APA style guidelines for accurate citation. Furthermore, data generated from interviews are regarded as intellectual outputs, jointly owned by the researcher and participants, and protected from unauthorized reproduction. Upholding these principles of intellectual property fosters trust, accountability, and ethical scholarship throughout this qualitative research endeavor.

B. Informed Consent

In qualitative research involving human participants, informed consent is an essential ethical consideration. It guarantees that individuals willingly participate, having received comprehensive information regarding the study's aims, methods, potential risks, and anticipated

benefits. For this study, in which PNP officers are participants, a written consent form will be provided, clearly outlining their rights, including the right to withdraw from the study at any time without consequence (Burkhardt et al., 2023). As Mantica et al., (2024) suggest, informed consent fosters transparency and mutual respect between researchers and participants by prioritizing voluntary involvement and protecting individual autonomy. Prior to each interview, participants will be informed about the confidentiality of their responses and assured that the data collected would be used exclusively for academic purposes. This procedure adheres to ethical research standards, which prioritize the dignity,respect, and freedom of choice of all participants.

C. Data Privacy and Confidentiality

To ensure the well-being of participants, this study strictly adheres to ethical standards and data privacy protocols, in compliance with the Data Privacy Act of 2012 (Republic Act No. 10173). All collected information will be securely stored. Entirely confidential and used solely for legitimate research purposes.

To protect the identities of Philippine National Police (PNP) officers, pseudonyms or special codes will be employed, ensuring their responses remain private. No published material will include any personally identifiable information. Participants will be fully informed of their rights before each interview, including the fact that participation is voluntary, their responses will be kept confidential, and their data will be handled securely and in accordance with applicable laws and regulations. Only the principal researcher and the thesis adviser will have access to password-protected folders containing digital files, such as audio recordings and transcripts. Physical records are in a secure location. All data will be retained only for the duration required by the study.

As highlighted by Shahid et al., (2022) maintaining confidentiality is essential to prevent the misuse or unauthorized disclosure of sensitive information that could potentially harm participants. By prioritizing data privacy, this study aims to foster trust with participants while upholding the integrity and credibility of the research process.

D. Voluntary Participation and Freedom to Withdraw

Voluntary participation, a fundamental ethical principle, guarantees individuals the freedom to decide whether or not to take part in a research study. In this qualitative research, police officers will participate in interviews freely, without any form of coercion, pressure, or incentive. As Donahue & Kleinberg (2021) suggest, ensuring voluntary participation helps to generate authentic responses, as individuals are more likely to provide honest insights when they engage willingly. Participants are explicitly informed that their decision to participate, or to withdraw at any time, will not affect their standing within the institution. This approach aligns with ethical guidelines designed to protect participants from undue influence or manipulation. By upholding voluntary participation, we demonstrate respect for individual autonomy and ensure that the collected data accurately reflects the participants' experiences and viewpoints.

E. Minimization of Harm and Risk Management

To minimize potential harm,this research incorporates measures to identify and mitigate any physical, emotional, or psychological risks to participants. Although the study is designed to pose minimal risk, we will take precautions to ensure interview questions do not cause discomfort or distress among participating PNP officers. Following Woods et al., (2022) we recognize the importance of anticipating potential emotional triggers and establishing support mechanisms

should participants experience unease. Participants will be assured of their right to skip any question or terminate the interview at any time should they feel uncomfortable. Our risk management strategies include debriefing participants after each session and adopting a neutral, non-judgmental tone throughout all discussions. By prioritizing participant well-being,this study upholds ethical responsibilities, ensuring the pursuit of knowledge does not compromise the dignity or emotional safety of respondents.Participants will be assured of their right to skip any question or terminate the interview at any time should they feel uncomfortable. Our risk management strategies include debriefing participants after each session and adopting a neutral, non-judgmental tone throughout all discussions. By prioritizing participant well-being, this study upholds ethical responsibilities, ensuring the pursuit of knowledge does not compromise the dignity or emotional safety of respondents.

F. Beneficence and Contribution to Knowledge

Beneficence, the ethical imperative to maximize benefits and minimize harms in research, was a guiding principle of this study. Our goal was to generate insights to inform policy and practice aimed at improving the experiences of PNP officers. As Steinert et al., (2021) emphasize, beneficence requires that research outcomes meaningfully contribute to society by addressing pertinent issues. We anticipate that the study's findings will assist personnel and policymakers in aligning programs to effectively address the challenges faced by PNP officers. Indirectly, participants may benefit from improvements to academic guidance and institutional support systems that result from this research. By upholding beneficence, we demonstrate a commitment to conducting research that not only protects participants but also contributes to the broader body of knowledge in the field.

G. Justice and Fair Participant Selection

In research, justice demands that participants are treated equitably and selected fairly,free from bias or discrimination. This study chose PNP officers based on their pertinence to the research question, avoiding selection based on convenience or personal preference. As O'Brien et al., (2022) articulated, justice in participant recruitment entails providing equal opportunities for involvement while safeguarding vulnerable populations from potential exploitation. All officers invited to participate will receive identical access to information, consent forms, and interview protocols. Furthermore,the principle of justice necessitates a fair distribution of research benefits, ensuring they are not confined to select individuals or groups. Maintaining fairness throughout the research process reinforces ethical accountability and strengthens the research's integrity by valuing all participants equally, irrespective of gender, socioeconomic status, or performance.

H. Data Integrity and Accuracy

Data integrity and accuracy represent the honest and meticulous gathering, analysis, and presentation of research results. In this study, all interview data will be transcribed verbatim and analyzed with the aim of objectivity,thereby mitigating distortion or bias. Consistent with Mutai (2024) assertion, data integrity is paramount in qualitative research, guaranteeing that interpretations faithfully represent participants' experiences,rather than reflecting the researcher's preconceived notions. To validate our findings and ensure consistency throughout the analysis, we will employ triangulation and peer review techniques. Crucially, we will not fabricate, alter, or selectively report data to sway the results. By upholding data integrity, we promote transparency, accountability, and scientific credibility, ensuring that the research conclusions are both reliable and ethically justifiable.

I. Transparency and Honesty in Reporting

To ensure transparency and honesty,this study adheres to rigorous reporting standards by presenting findings truthfully, without fabrication,falsification, or selective omission. All results, interpretations, and limitations are clearly articulated to provide an accurate representation of participant perspectives. As Ramesh (2024) notes,research transparency is crucial for fostering public trust and ensuring ethical knowledge production. Any potential conflicts of interest are disclosed, and the implications of the findings are carefully considered to avoid overstatement. Furthermore, comprehensive methodological details are provided to facilitate replication and extension of this work by future researchers. By upholding honesty in reporting, we reinforce research integrity and demonstrate respect for both participants and the broader research community, ensuring the authenticity and verifiability of the knowledge generated.

J. Use of Patented or Copyrighted Materials

When incorporating existing research into our work, we are committed to upholding intellectual property rights and following ethical guidelines for attribution. This study adheres to APA referencing standards for all published materials, including books, journal articles, and electronic resources. As Lalwani (2025) highlights,reproducing or altering copyrighted material without authorization compromises academic integrity and can have legal consequences. Any third-party data,figures, or instruments used in this research will be either publicly available or employed with the explicit permission of the original creators. By respecting intellectual property, we not only protect the rights of authors but also demonstrate ethical research practices for others to follow. This commitment to intellectual ownership reflects our dedication to honesty,fairness, and respect for the work of fellow researchers.

K. Consideration for Human Subjects

Throughout the research process,respect for the dignity,rights, and well-being of participants will be paramount. In this qualitative study involving PNP officers, all interactions will be characterized by sensitivity, empathy, and professionalism. Following Buchanan & Warwick (2021) we recognize participants not simply as data sources but as active contributors, prioritizing their protection from exploitation, distress, or any form of harm. To foster a safe and respectful interview environment,the researcher will remain culturally sensitive and adapt communication styles as needed. Furthermore, participants will be assured that their perspectives will be accurately reflected in the study's findings. By prioritizing consideration for human subjects,this research aims to advance academic knowledge while simultaneously promoting ethical, humane, and socially responsible inquiry.

L. Responsible Use of AI and Other Technologies

This research prioritizes the responsible integration of AI and other technologies,focusing on transparency, accountability, and integrity in data handling and the generation of findings. Technological tools, including transcription software, data management systems, and AI-assisted analysis, will be used specifically to enhance accuracy and efficiency, while strictly adhering to ethical guidelines. Aligning with Christou (2023) emphasis,these tools will serve as aids, supporting rather than replacing critical human interpretation,to minimize bias and prevent misrepresentation of qualitative data. Moreover, all technological applications will comply with data privacy regulations, notably the Data Privacy Act of 2012, ensuring the confidentiality and secure storage of participant data. To maintain transparency, any use of AIassisted systems in data

processing or writing will be clearly disclosed. This responsible integration of technology allows the study to embrace innovation while upholding ethical accountability in research practices.

M. Ethical Clearance and Institutional Approval

Prior to any data collection,this research obtained both ethical clearance and institutional approval, vital steps to ensure adherence to ethical standards and the protection of participant well-being. Formal approval was secured from the relevant ethics review committee or institutional authority, demonstrating a commitment to established ethical protocols and regulatory guidelines. As highlighted by Ehidiamen et al., (2024) obtaining ethical clearance confirms that the study's aims, methods, and informed consent procedures respect human dignity and align with institutional and national ethical frameworks. This process prioritizes the rights, safety, and confidentiality of participants throughout the study. Furthermore, institutional approval underscores the researcher's commitment to accountability, scholarly integrity, and compliance with educational and ethical requirements. Through this dual process of ethical clearance and institutional oversight,the study reinforces its credibility and dedication to responsible academic inquiry.

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